SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.4: Chemical Bonding — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.4: Chemical Bonding
Driving Question	DRIVING QUESTION: Why do diamond and graphite — both made of only carbon atoms — have such completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Carbon Paradox: Launching the Phenomenon**	Students handled or viewed images of diamond (transparent, extremely hard, used in Nairobi jewellery shops and industrial drill bits) and graphite (grey, soft, slippery, used in pencil 'leads'). They recorded contrasting physical properties — hardness, colour, texture, conductivity — and noted the paradox that both substances are made of only carbon atoms.	Diamond and graphite are allotropes of carbon: same element, same atomic number (6), yet dramatically different physical properties. The difference cannot come from the atoms themselves — it must come from HOW those atoms are connected to each other. This launches the sub-strand's central question: bond arrangement, not atomic identity, governs a substance's properties.	Teacher note: This lesson is intentionally light on answers — its purpose is to generate productive confusion and anchor all future learning to a concrete, memorable phenomenon. Resist the urge to explain bonding yet. Instead, cold-call 3–4 students to articulate what is puzzling them and record their questions on the Driving Question Board. The phrase 'same atoms, different arrangement' should be the only explanatory seed planted here.
Lesson 2 **Valence Electrons & the Octet Rule: The Electron Bookkeeping Behind the Carbon Paradox**	Students used period-table period numbers and group numbers to determine valence electron counts for H, C, N, O, Na, Mg, Cl, and Ar. They drew dot-and-cross (Lewis dot) diagrams for individual atoms and observed that noble gases (full outer shells) are chemically inert, while other atoms have incomplete outer shells.	Valence electrons — those in the outermost energy level — are the only electrons involved in bonding. Atoms bond in order to achieve a full outer shell: 8 electrons (the octet rule) for most elements, or 2 electrons (duplet rule) for hydrogen and helium. Carbon has exactly 4 valence electrons, giving it unique flexibility: it can form 4 bonds in multiple geometric arrangements — the root cause of the diamond–graphite paradox.	Teacher note: Emphasise that the octet rule is a useful model, not an absolute law (exceptions such as PCl_5 exist but are beyond Grade 10 scope). Use the carbon atom as the recurring anchor: ask students, 'If carbon has 4 valence electrons and needs 4 more to complete its octet, what are its bonding options?' Allow 30 seconds of wait time before cold-calling. This question should remain open and be revisited in Lessons 3, 4, and 10.
Lesson 3 **Ionic Bonding: Electron**	Students examined the properties of sodium chloride (table salt — familiar from	Ionic bonds form when a metal atom transfers one or more valence electrons to	Teacher note: Contrast ionic bonding explicitly with what students will learn about

Transfer and the Giant Ionic Structure	salting ugali and nyama choma): white crystalline solid, high melting point, brittle when struck, conducts electricity only when dissolved in water or melted. They also observed the flame test colours for Na^+ (yellow) and the vigorous reaction of sodium metal with water (teacher demonstration or video).	a non-metal atom, forming oppositely charged ions held together by strong electrostatic attraction. NaCl forms a giant ionic lattice — a three-dimensional array of alternating Na^+ and Cl^- ions. This lattice structure explains NaCl's high melting point (strong forces between many ions), brittleness (ion layers shift and like charges repel), and conductivity only in solution or melt (ions must be free to move). Ionic bonding is distinct from covalent bonding because electrons are transferred, not shared.	carbon in Lessons 4 and 10. Ask: 'Carbon does NOT form ionic bonds in diamond or graphite — why not?' (Carbon is a non-metal and has equal electronegativity to itself, so electron transfer does not occur.) This keeps the driving question alive. Use dot-and-cross diagrams to show electron transfer for NaCl and MgO. Cold-call 2 students to explain why ionic compounds do not conduct in the solid state using particle-level reasoning.
Lesson 4 **Covalent & Dative Covalent Bonding: Drawing Lewis Structures and Exploring Electron Sharing**	Students drew Lewis (dot-and-cross) structures for H_2, Cl_2, H_2O, NH_3, CO_2, and CH_4. They observed that carbon in CH_4 forms exactly 4 shared pairs, achieving a full octet. They also examined the dative (coordinate) bond in NH_4^+, where nitrogen donates both electrons of a lone pair to form a bond with H^+.	Covalent bonds form when two atoms share one or more pairs of electrons, with each atom typically contributing one electron to the shared pair. A dative (coordinate) covalent bond is a special case where one atom donates BOTH electrons of the shared pair. Bond order (single, double, triple) increases bond strength and decreases bond length. Carbon's 4 valence electrons allow it to form 4 covalent bonds — this flexibility is the foundation for understanding why diamond and graphite have different structures: both use covalent bonds, but the geometry and number of bonds per atom differ.	Teacher note: The critical pedagogical move here is to withhold the full diamond/graphite explanation. Students now know carbon forms 4 covalent bonds — but HOW those bonds are arranged in 3-D space is the key, which is addressed in Lessons 10 and 12. Pose the question: 'Carbon forms 4 covalent bonds in both diamond and graphite — so why are they different?' and add it to the Driving Question Board. Allow students to make predictions. This productive uncertainty deepens engagement with Lesson 10.
Lesson 5 **Hydrogen Bonding & Van der Waals Forces: Why Water Defies Expectation**	Students compared the boiling points of H_2O (100 °C), H_2S (−60 °C), HCl (−85 °C), and CH_4 (−161 °C). They noted that water has an anomalously high boiling point relative to its molar mass. They also observed water's surface tension (a paper clip floating on water in a beaker — a familiar sight at Kenyan household water containers) and water's ability to dissolve ionic compounds like NaCl.	Intermolecular forces are attractions between molecules, distinct from the covalent bonds within molecules. Van der Waals forces (London dispersion forces and permanent dipole–dipole interactions) are weak, temporary or permanent electrostatic attractions present in all molecules; they increase with molecular mass and surface area. Hydrogen bonds are a stronger type of intermolecular force that form when hydrogen is covalently bonded to a highly electronegative atom (N, O, or F) and is attracted to a lone pair on a neighbouring N, O, or F. Water's two O–H bonds and two lone pairs allow it to form an extensive hydrogen-bond network, explaining its high boiling point, high specific heat capacity, and role as the 'solvent of life' — critical for	Teacher note: Emphasise that hydrogen bonds are intermolecular (between molecules), NOT intramolecular (within a molecule) — this is a persistent student misconception. Use a think-pair-share: 'Why does water have a higher boiling point than hydrogen sulfide, even though H_2S has a greater molar mass?' Allow 60 seconds pair discussion before cold-calling. Connect to daily Kenyan life: the fact that uji (porridge) takes a long time to cool is partly due to water's high specific heat capacity, itself a consequence of hydrogen bonding.

		understanding Lake Victoria's climate-moderating effect.	
Lesson 6 **Metallic Bonding: The Sea of Electrons**	Students examined physical properties of metals available in the Kenyan context — iron (from a jua kali workshop tool), copper wire (from electrical wiring), aluminium foil (used to wrap nyama choma). They recorded that metals are lustrous, malleable, ductile, good conductors of heat and electricity, and generally have high melting points.	Metallic bonds form in solid metals when metal atoms release their valence electrons into a shared, delocalised 'sea of electrons' that surrounds a lattice of positive metal ions (cations). The strong electrostatic attraction between the positive ion lattice and the mobile electron sea accounts for metals' characteristic properties: high melting points (strong attraction throughout the lattice), electrical conductivity (free electrons carry charge), thermal conductivity (free electrons transfer kinetic energy), malleability and ductility (ion layers slide without breaking the non-directional metallic bond), and metallic lustre (free electrons absorb and re-emit light). Metallic bonding is a third distinct bond type alongside ionic and covalent — and like giant ionic and giant covalent structures, the giant metallic structure involves a vast 3-D lattice held together by strong forces throughout.	Teacher note: This lesson completes the trio of primary bond types (ionic, covalent, metallic) before students investigate physical properties experimentally in Lessons 7–8. Explicitly ask students to compare and contrast the 'sea of electrons' model with the ionic and covalent models they already know. A useful cold-call prompt: 'A copper wire conducts electricity — a giant ionic lattice of NaCl does not — use particle-level reasoning to explain why.' Wait 45 seconds before taking responses. Add to the Driving Question Board: 'How does the sea of electrons explain why we can bend a copper wire but not a salt crystal?'
Lesson 7 **Physical Properties Investigation — Part 1: Setup & Prediction**	Students were introduced to four substances to be investigated: NaCl (giant ionic), candle wax (simple molecular), sand/SiO_2 (giant covalent), and copper wire (giant metallic). They recorded initial observations (appearance, feel, colour) and made written predictions about each substance's melting point range (low/medium/high), solubility in water, and electrical conductivity in solid and dissolved states, justifying each prediction using their knowledge of bond types from Lessons 2–6.	Each structural type — giant ionic, simple molecular, giant covalent, and giant metallic — has a characteristic pattern of physical properties that can be predicted from the nature and strength of the forces that must be overcome. Giant structures (ionic, covalent, metallic) all require breaking many strong bonds throughout a lattice and therefore have high melting points; simple molecular substances require only overcoming weak intermolecular forces and therefore have low melting points. Conductivity depends on the presence of freely moving charged particles — ions or electrons. Making and justifying predictions before collecting data is a core scientific practice.	Teacher note: This is a prediction-only lesson — data collection occurs in Lesson 8. The pedagogical value lies in students committing to written predictions with reasoning BEFORE they see results. This creates cognitive investment and makes discrepancies between prediction and observation more memorable. Circulate and check prediction tables; look for students who conflate 'melting the solid' with 'dissolving in water' for ionic compounds — this is a common misconception to address in Lesson 8. Do NOT correct predictions at this stage; allow the experiment to do that work.
Lesson 8 **Physical Properties Investigation — Part 2: Data Collection**	Students tested the four substances (NaCl, candle wax, SiO_2/sand, copper wire) for: (1) solubility in water, (2) electrical conductivity as a solid and in solution/melt using a	Bond type determines physical properties in predictable ways: ionic compounds (NaCl) conduct only when dissolved or melted because ions become free to move; in the	Teacher note: After data collection, run a 'claim–evidence–reasoning' (CER) discussion: select 2–3 student groups to present one result each, explicitly naming

(Solubility, Conductivity & Melting Point)	simple circuit with a bulb or LED, and (3) relative melting behaviour using a hot-water bath or candle flame. They recorded results in a data table and compared findings with their Lesson 7 predictions.	solid state, ions are fixed in the lattice. Simple molecular substances (candle wax) have low melting points because only weak van der Waals forces between molecules must be overcome; covalent bonds within molecules are not broken on melting. Giant covalent structures (SiO_2/sand) have very high melting points because every atom is held by strong covalent bonds throughout the lattice. Giant metallic structures (copper) conduct in both solid and liquid states because delocalised electrons are always free to move. These empirical patterns confirm and extend the theoretical models built in Lessons 3–6.	the claim, their evidence, and the reasoning that connects evidence to claim. Pay particular attention to NaCl conductivity — many students predict it will conduct as a solid. Use this as a teachable moment: ask 'What would the ions need to do in order to carry charge, and can they do that in a rigid lattice?' Connect back to the driving question: 'We now have a tool — bond type predicts properties. How does this help us start explaining diamond vs. graphite?'
Lesson 9 **Giant Covalent Structures: Diamond, Graphite, and Silicon Dioxide in Depth**	Students examined diagrams and, where available, physical samples or 3-D model images of diamond, graphite, and SiO_2 (quartz/sand). They annotated diagrams to show bond angles (109.5° tetrahedral in diamond; 120° trigonal planar within graphite layers), the number of covalent bonds per carbon atom in each allotrope (4 in diamond; 3 in graphite, with 1 delocalised electron per atom), and the presence or absence of free electrons.	Diamond, graphite, and SiO_2 are all giant covalent structures — three-dimensional networks in which every atom is linked to its neighbours by strong covalent bonds — which is why they all have very high melting points and are insoluble in water. However, their internal geometries differ critically: in diamond, each carbon forms 4 tetrahedral covalent bonds to 4 neighbours, creating a rigid, isotropic 3-D lattice with no free electrons (hence non-conductive, hardest natural substance). In graphite, each carbon forms 3 bonds in a flat hexagonal layer; the 4th valence electron is delocalised across the layer (hence conducts electricity within layers). The weak van der Waals forces between graphite layers allow them to slide (hence soft, lubricating, and used in pencil leads). In SiO_2, each Si forms 4 bonds to O atoms — similar geometry to diamond, explaining its hardness and high melting point but non-conductivity.	Teacher note: This is the conceptual resolution lesson for the driving question — students now have the full answer to the carbon paradox. Use a direct 'return to the phenomenon' moment: display the diamond and graphite images from Lesson 1 side by side and ask students to write a full explanation in their own words before class discussion. Cold-call 2 students to give their explanations; use peer critique ('What did they include? What could be added?'). Emphasise that both allotropes are 'giant covalent' but the 3-D geometry of bonds — not the bond type alone — determines the property differences. This is the sub-strand's central learning breakthrough.
Lesson 10 **Resultant Structures & Uses: From Bond Type to Real-World Application**	Students matched real-world materials and applications to their bond types and structural features: diamond (industrial drill bits used in Kenyan mining operations, gemstones), graphite (pencil leads, dry lubricants, electrodes in electrolysis), SiO_2 (glass, optical fibres for Nairobi's internet infrastructure, ceramic tiles), NaCl (food	A material's real-world application is directly determined by its bond type and resultant structure: diamond's rigid tetrahedral lattice (no free electrons, no cleavage planes) makes it ideal for cutting and drilling; graphite's layered structure with delocalised electrons makes it an excellent lubricant and electrode; SiO_2's giant covalent	Teacher note: This lesson is the 'payoff' for all prior content and should feel integrative and satisfying. A recommended activity is a 'Design a Material' challenge: give student pairs a real Kenyan engineering problem (e.g., 'A Rift Valley geothermal plant needs a material for high-temperature drill bits — what bond type and structure do you

	preservation of fish at Lake Victoria, saline drips in hospitals), copper (electrical wiring throughout Kenya's national grid), and simple molecular substances (iodine as antiseptic, candle wax, plastics).	network gives glass and optical fibres their transparency and durability; NaCl's giant ionic lattice dissolves readily in water, enabling its use in food preservation and medicine; copper's metallic sea of electrons provides the conductivity and malleability needed for electrical wiring. The principle 'structure determines properties; properties determine use' is the master organising idea of the sub-strand.	recommend, and why?') and have them justify their answer in writing. This develops both higher-order thinking and real-world relevance. Add the completed 'structure → properties → use' chain to the Driving Question Board as the sub-strand's master answer.
Lesson 11 **Animations, Simulations & Digital Literacy: Seeing Chemical Bonding in Motion**	Students used digital simulations (PhET 'Build a Molecule', ARES platform animations, or teacher-projected 3-D models) to visualise the tetrahedral network of diamond, the layered hexagonal lattice of graphite, the ionic lattice of NaCl, and the electron-sea model of copper. They annotated screenshots or sketches with labels identifying bond type, geometry, and the structural feature that explains a key property.	Digital simulations and animations allow students to 'see' structural features that are impossible to observe directly — bond angles, lattice repeating units, electron delocalisation, and layer separation. Critically evaluating a simulation (identifying what it accurately models and what it simplifies or omits) is a scientific and digital literacy skill. For example, 3-D models of graphite show layers as perfectly flat and equidistant, whereas real graphite has slight interlayer disorder — the simulation is a useful approximation, not a perfect replica. Students practised cross-referencing simulation outputs with data tables from Lesson 8 to check for consistency.	Teacher note: For schools without reliable internet, pre-downloaded ARES platform animations or printed 3-D diagram sheets can replace live simulations. The key pedagogical goal is not technology use per se but the habit of triangulating multiple representations — diagram, simulation, physical data, and written explanation — as practised scientists do. Ask students: 'What does this simulation help you understand that a 2-D diagram cannot show?' Cold-call 2 students. Also prompt: 'What does the simulation NOT show that you would need to know to fully explain graphite's conductivity?' (Answer: it does not show electron movement.)
Lesson 12 **Project Lesson — 3-D Model Building: Bonding Structures with Locally Available Materials**	Students constructed physical 3-D models of assigned bonding structures (diamond, graphite, NaCl ionic lattice, or a simple molecular substance such as CH_4 or H_2O) using locally available materials: maize cobs or mandazi pieces as atoms, matchsticks or toothpicks as bonds, wire or string for metallic/delocalised electron representation. They measured or estimated bond angles in their models and presented them to the class.	The three-dimensional geometry of chemical bonds — 109.5° tetrahedral in diamond and SiO_2, 120° trigonal planar within graphite layers — cannot be fully appreciated from 2-D diagrams alone. Building a physical model makes bond angles, coordination numbers, and spatial relationships tangible and memorable. Students also learned that all models are simplifications: a maize-cob-and-matchstick model cannot show electron density, bond polarity, or thermal vibration, and acknowledging a model's limitations is part of scientific modelling practice. Cross-group comparison of diamond and graphite models provided a final visual confirmation of the driving question's answer.	Teacher note: Assign structures strategically — ensure at least two groups build diamond and two build graphite so a direct side-by-side comparison is possible during presentations. During the gallery walk or presentation phase, require each group to state: (1) the bond type in their structure, (2) one physical property their model helps explain, and (3) one real-world Kenyan use of the substance. This ties project work back to sub-strand learning goals. Assess models formatively using a simple three-point rubric: accuracy of geometry, correctness of bond type representation, and quality of oral explanation.
Lesson 13 **Sub-Strand Review &**	Students revisited the Driving Question Board compiled across all 12 previous	All major concepts of Sub-Strand 1.4 are now integrated into a coherent explanatory	Teacher note: This lesson serves as both consolidation and formative/summative

Assessment: Consolidating the Carbon Paradox	lessons, reviewing how each posted question was eventually answered. They completed a structured sub-strand review activity — matching bond types to properties, explaining the diamond–graphite paradox in full written sentences, and evaluating a novel material (e.g., silicon carbide used in Kenyan solar panel components) by predicting its properties from its giant covalent structure.	framework: bond type (ionic, covalent — including dative, hydrogen bonding, van der Waals, metallic) determines the resultant structure (giant ionic, simple molecular, giant covalent, giant metallic), which in turn determines physical properties (melting point, solubility, conductivity, hardness), which in turn determines real-world applications. The diamond–graphite paradox is fully explained: same atoms (carbon), same bond type (covalent), but different 3-D geometry (4 bonds tetrahedral vs. 3 bonds trigonal planar with delocalised electrons) produces entirely different properties and uses.	assessment. The Driving Question Board review is a powerful metacognitive tool — walk through it chronologically with the class, asking 'Which lesson answered this question?' to make learning progression visible. For the novel-material transfer task (silicon carbide), do NOT pre-teach — the ability to apply the 'structure → properties → use' framework to an unfamiliar substance is the target competency for Grade 10 CBE. Collect written explanations as an assessment artefact. Celebrate students who can articulate that the sub-strand's answer is not just about carbon, but about a universal principle: arrangement of bonds determines the identity of a material.